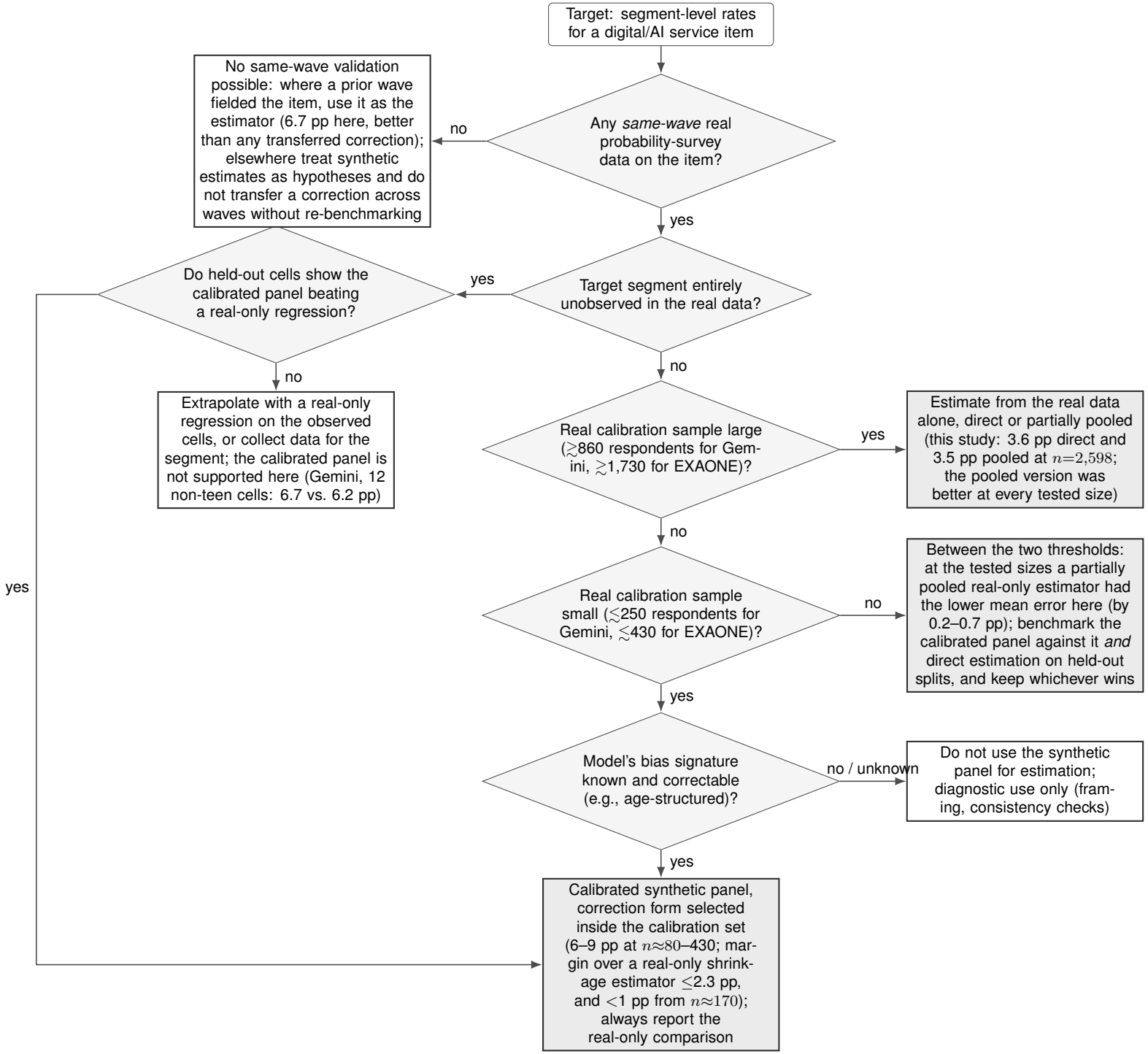

Target: segment-level rates for a digital/AI service item
Any *same-wave* real probability-survey data on the item?
no
No same-wave validation possible: where a prior wave fielded the item, use it as the estimator (6.7 pp here, better than any transferred correction); elsewhere treat synthetic estimates as hypotheses and do not transfer a correction across waves without re-benchmarking
yes
Target segment entirely unobserved in the real data?
yes
Do held-out cells show the calibrated panel beating a real-only regression?
no
Extrapolate with a real-only regression on the observed cells, or collect data for the segment; the calibrated panel is not supported here (Gemini, 12 non-teen cells: 6.7 vs. 6.2 pp)
yes
no
Real calibration sample large (≳860 respondents for Gemini, ≳1,730 for EXAONE)?
yes
Estimate from the real data alone, direct or partially pooled (this study: 3.6 pp direct and 3.5 pp pooled at $n{=}2{,}598$; the pooled version was better at every tested size)
no
Real calibration sample small (≲250 respondents for Gemini, ≲430 for EXAONE)?
no
Between the two thresholds: at the tested sizes a partially pooled real-only estimator had the lower mean error here (by 0.2–0.7 pp); benchmark the calibrated panel against it *and* direct estimation on held-out splits, and keep whichever wins
yes
Model's bias signature known and correctable (e.g., age-structured)?
no / unknown
Do not use the synthetic panel for estimation; diagnostic use only (framing, consistency checks)
yes
Calibrated synthetic panel, correction form selected inside the calibration set (6–9 pp at $n{\approx}80$–430; margin over a real-only shrinkage estimator ≤2.3 pp, and <1 pp from $n{\approx}170$); always report the real-only comparison